

Hardware Integration

1. Overview

Our architecture is a **closed loop** where the receiver and RL algorithm remain independent of the specific hardware. Only **two interface layers** connect them: **one converts receiver data → standard observation**, and the **other converts RL decision → receiver command**. This means the receiver or its communication method can be changed without rewriting the RL algorithm. This is our **hardware-independent design** choice, not a claim that one universal protocol exists.

2. Receiver + Data Interface

The receiver **detects signals/pulses** and produces measurements, such as time, frequency, pulse width, signal strength and, where available, direction information. These are represented as a PDW (Pulse Descriptor Word). The exact way this information reaches our software depends on the receiver: it could use **VITA-49 or another vendor-specific interface**. Our **Data Adapter** converts whatever the receiver provides into one common observation format for the scheduler.

(Here we can define the Receiver → Scheduler data contract => exactly what fields/format the scheduler receives after the receiver interface)

3. Smart Scheduler

The scheduler only **sees these observations**, never the hidden truth or receiver-specific protocol. It **uses its scan history and the RL model to decide which band to observe next** and, if supported, for how long. Its output is a simple hardware-independent decision such as “Band 12, dwell 50 ms”, not a vendor command. This is what keeps the intelligence portable.

4. Tuning / Control Interface

A **Control Adapter** takes the scheduler's simple decision and converts it into whatever command the actual receiver understands. Depending on the receiver, that could be a vendor API, SCPI, or a supported VITA-49.2 control mechanism. The **receiver executes the command and produces the next observation, completing the loop**.

(Here too we can define the Scheduler → Receiver command contract => what fields and format our tuning command contains)

For example {frequency, bandwidth, dwell_time} → Control Adapter → receiver-specific command.

Closed-Loop Architecture

Intelligent Spectrum Scanning

